

EX-07 Development of a Prompt-Based Application using Large Language Models

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Aim

To demonstrate the development of a prompt-based application using Large Language Models (LLMs) for organizing daily tasks and improving productivity through effective prompt engineering techniques.

Objective

To create a Daily Productivity Assistant using AI tools such as OpenAI ChatGPT that helps users plan tasks, manage time, and improve productivity using simple and refined prompts.

Introduction

Large Language Models (LLMs) can be used to create intelligent applications through prompt engineering. In this experiment, a Daily Productivity Assistant application is developed using prompts that guide the AI to generate personalized schedules and productivity suggestions. The experiment demonstrates how prompt refinement improves output quality and user experience.



Problem Statement

Many students struggle to manage their daily schedules efficiently due to poor planning and distractions. A prompt-based productivity assistant can help organize tasks and allocate study time effectively.

Selected Application

Daily Productivity Assistant

Objective of the Application

- Organize daily activities
- Improve time management
- Increase productivity
- Reduce task overload

- Generate personalized schedules

Algorithm

Step 1: Identify the Problem

- Select a daily-life productivity problem.
- Define the objective of improving daily task management.

Step 2: Design Basic Prompt

- Create a simple productivity prompt.
- Generate initial AI output.

Step 3: Refine the Prompt

- Add constraints, priorities, and formatting instructions.
- Improve prompt clarity.

Step 4: Generate Advanced Outputs

- Generate personalized schedules.
- Compare simple and refined outputs.

Step 5: Analyze the Results

- Evaluate effectiveness of prompts.
- Compare output quality and usefulness.

Step 6: Document the Application

- Include prompts, outputs, screenshots, observations, and comparison tables.

Step 1 – Basic Prompt

Simple Prompt

“Create a daily study schedule for a college student.”

Output Generated by ChatGPT

Time	Activity
6:00 AM – 7:00 AM	Exercise
7:00 AM – 8:00 AM	Breakfast and preparation
9:00 AM – 1:00 PM	College classes
2:00 PM – 4:00 PM	Study session
5:00 PM – 6:00 PM	Break
7:00 PM – 9:00 PM	Homework and revision
10:00 PM	Sleep

Limitations of Basic Prompt

- No personalization
- No subject prioritization
- No break optimization
- Generic schedule
- Less detailed output

Step 2 – Refined Prompt

Refined Prompt

“You are a productivity assistant for an engineering student studying Artificial Intelligence and Data Science. Create a detailed daily schedule including study time, breaks, coding practice, exercise, revision, and sleep. Prioritize programming subjects and include productivity tips. Present the output in a clear table format.”

Refined Output Generated by ChatGPT

Time	Task	Description
5:30 AM – 6:00 AM	Morning Routine	Freshen up and meditation
6:00 AM – 7:00 AM	Exercise	Walking and stretching
7:00 AM – 8:00 AM	Breakfast	Healthy breakfast
8:00 AM – 1:00 PM	College Classes	Attend lectures and lab sessions
1:00 PM – 2:00 PM	Lunch Break	Relax and refresh
2:00 PM – 4:00 PM	Coding Practice	Practice C++ and Python programming
4:00 PM – 4:30 PM	Short Break	Snacks and relaxation
4:30 PM – 6:00 PM	AI & Data Science Study	Learn AI concepts and notes
6:00 PM – 7:00 PM	Recreation	Music or social activities
7:00 PM – 8:30 PM	Assignment Work	Complete college assignments
8:30 PM – 9:00 PM	Dinner	Healthy meal
9:00 PM – 10:00 PM	Revision	Review class topics
10:30 PM	Sleep	Proper rest

Advanced Prompt Design Technique Used

Role-Based Prompting

The AI acts as a productivity assistant.

Constraint-Based Prompting

Specific instructions were given regarding:

- Study hours
- Programming priority
- Breaks

- Formatting style

Personalized Prompting

The schedule was customized for an AI & Data Science student.

Comparison Between Basic and Refined Prompt

Feature	Basic Prompt	Refined Prompt
Detail Level	Low	High
Personalization	No	Yes
Formatting	Simple	Structured
Productivity Tips	No	Included
Subject Priority	No	Yes
User Satisfaction	Moderate	High

Observations

- Refined prompts generated more useful and organized outputs.
- Personalized prompts improved relevance.
- Role-based prompting increased contextual understanding.
- Detailed instructions improved output clarity and structure.
- AI-generated schedules became more practical after refinement.

Advantages of Prompt-Based Applications

- Saves time
- Improves productivity

- Generates personalized solutions
- Easy to modify prompts
- Provides structured outputs
- Useful for students and professionals

Output

The Daily Productivity Assistant successfully generated:

- Personalized schedules
- Structured task planning
- Study management strategies
- Improved productivity suggestions

Result

Thus, a prompt-based application was successfully developed using Large Language Models to solve daily productivity and scheduling problems. The experiment demonstrated that refined prompts produce more accurate, structured, and personalized outputs compared to simple prompts.

Conclusion

This experiment showed that prompt engineering plays a major role in improving AI-generated responses. Refined prompts with role-based and constraint-based instructions produced better-quality outputs than simple prompts. Prompt-based applications can effectively assist users in solving practical daily-life problems.

References

1. [OpenAI ChatGPT](#)
2. [Prompt Engineering Guide](#)
3. [OpenAI Documentation](#)

